Statistical Mechanics of Money:

Applying Boltzmann-Gibbs Distributions and Entropy
to Economic Dynamics via Agent-Based Simulation

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Contents

Abstract

Keywords: Statistical mechanics, Boltzmann-Gibbs distribution, Gibbs entropy, Agent-based simulation, Markov processes, Economic dynamics, Complex systems.

1 Introduction

The applicability of **statistical mechanics** to economic systems, specifically how money distribution in a society of interacting economic agents follows the **Boltzmann-Gibbs distribution**. [1, 2] This is a foundational principle from thermodynamics and statistical physics. The assumption is that money, like energy in a physical system, is conserved during transactions between agents.

I conducted agent based simulations modeling economic transactions as random exchanges of money between agents, analogous to particle collisions in a gas. Under various transaction mechanisms such as constant exchanges between agents, fraction γ of individual wealth per agent in each transaction, fraction of system average, etc., the simulated money distribution converges to an exponential form $P(m) = \frac{1}{T}e^{-m/T}$, where $T = M/N$ represents the system's effective "temperature" (average wealth per agent).

Two simulation models are analyzed:

1. **Basic money distribution** with no debt constraints

2. **System with debt** allowing limited negative balances

It's possible to demonstrate that **Gibbs entropy** correctly predicts equilibrium states and that the system evolves as a **Markov process** toward maximum entropy. These results validate the hypothesis that economic inequality and wealth concentration emerge as natural equilibrium states when agents interact via random transactions, analogously to how physical systems spontaneously approach thermodynamic equilibrium.

This research connects fundamental concepts from statistical mechanics from Markov processes, Boltzmann-Gibbs distributions, Gibbs entropy, and dynamical systems theory to economics and the so called "econophysics" or "social physics", providing a computational validation of theoretical predictions.

1.1 Motivation

The term "econophysics" was introduced by analogy with similar terms, such as astrophysics, geophysics, and biophysics, which describe applications of physics to different fields.

It should be emphasized that econophysics does not literally apply the laws of physics, such as Newton's laws or quantum mechanics, to humans, but rather uses mathematical methods developed in statistical physics to study properties of complex economic systems consisting of a large number of humans. So, it may be considered as a branch of applied theory of probabilities. Econophysics distances itself from the verbose, narrative, and ideological style of political economy and is closer to econometrics in its focus. Studying mathematical models of a large number of interacting economic agents, econophysics has much common ground with the agent-based modeling and simulation [3].

1.2 Boltzmann-Gibbs Distribution of energy

The fundamental law of equilibrium statistical mechanics is the Boltzmann–Gibbs distribution. It states that the probability $P(\varepsilon)$ of finding a physical system or subsystem in a state with the energy ε is given by the exponential function

$$P(\varepsilon) = ce^{-\varepsilon/T}, \tag{1}$$

where T is the temperature, and c is a normalizing constant. Here we set the Boltzmann constant k_B to unity by choosing the energy units for measuring the physical temperature T. Then, the expectation value of any physical variable x can be obtained as

$$\langle x \rangle = \frac{\sum_k x_k e^{-\varepsilon_k/T}}{\sum_k e^{-\varepsilon_k/T}}, \tag{2}$$

where the sum is taken over all states of the system. Temperature is equal to the average energy per particle: $T \sim \langle \varepsilon \rangle$, up to a numerical coefficient of the order of 1.

Equation (1) can be derived in different way. One of the shortest derivations can be summarized as follows. Let us divide the system into two (generally unequal) parts. Then, the total energy is the sum of the parts: $\varepsilon = \varepsilon_1 + \varepsilon_2$, whereas the probability is the product of probabilities: $P(\varepsilon) = P(\varepsilon_1)P(\varepsilon_2)$. The only solution of these two equations is the exponential function (1).

2 Model and Theory

2.1 Theoretical Framework

[Model and theory content to be filled...]

3 Results

3.1 Scenario 1: Basic Money Distribution

[Results content to be filled with figures and analysis...]

4 Conclusion

4.1 Summary

[Conclusion content to be filled...]

A Mathematical Details

A.1 Boltzmann-Gibbs Distribution

[Mathematical details to be filled...]

B Code Snippets

B.1 Core Simulation

[Code snippets to be filled...]

References

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[3] Victor M. Yakovenko and J. Barkley Rosser. Colloquium: Statistical mechanics of money, wealth, and income. *Reviews of Modern Physics*, 81(4):1703–1725, 2009.